

Udit Agarwal

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EDUCATION

Indiana University Bloomington

Master of Science in Computer Science, GPA: 3.84/4.0

Bloomington, Indiana, US

Aug 2024 - May 2026

Kalinga Institute of Industrial Technology (KIIT University)

Bachelor of Technology in Computer Science and Engineering, GPA: 8.85/10.0

Bhubaneswar, Odisha, India

Aug 2020 - May 2024

EXPERIENCE

Heartland Community Network

Senior Consultant

Jun 2026 - Present

Remote, Indiana, US

- Architect and deliver full-stack applications, backend services, and REST APIs against client-specific operational requirements, owning design, implementation, testing, deployment, and ongoing maintenance
- Engineer AI-powered features and LLM-driven automation workflows that cut manual process steps, speed up information retrieval, and improve data-driven decision-making across client industries
- Translate complex and ambiguous client requirements into scalable technical designs, then own delivery through code review, testing, deployment, and maintenance

Global Health Impact Project

Software Engineer

Jan 2026 - Jun 2026

Indiana University, Bloomington, Indiana

- Engineered full-stack analytics features in React, TypeScript, and Python enabling researchers to evaluate pharmaceutical intervention coverage, efficacy, and population-level outcomes
- Built backend services and data pipelines for treatment-coverage modeling, efficacy analysis, and large-scale population health forecasting, exposed through REST APIs to forecasting dashboards
- Identified and resolved application bottlenecks and production issues across the stack, optimizing database queries and backend processing to support increasingly complex analytical workloads

University Information Technology Services (UITS)

Consultant - IT Services

Aug 2025 - Present

Indiana University, Bloomington, Indiana

- Resolve 100+ technical issues weekly across desktop, mobile, and enterprise applications through structured troubleshooting and root-cause analysis, documenting resolutions in ServiceNow
- Support enterprise identity and access management across authentication, authorization, account provisioning, and lifecycle workflows for a large university user base

PROJECTS

PrepWise - AI Voice Interview Platform | *Next.js, TypeScript, Firebase, Web Speech API, Monaco Editor, Pyodide/WASM, Gemini, Zod*

- Shipped live coding rounds with real in-browser execution for JavaScript and Python, killing infinite loops at a 5-second (JS) and 15-second (Python) timeout without freezing the tab, by running code in a terminable Web Worker with a lazily loaded Pyodide WASM runtime
- Cut per-interview speech cost to zero across 5 interview formats by replacing paid speech services with browser-native recognition and synthesis, rebuilding capture as continuous recognition that accumulates segments across pauses
- Kept the progress dashboard to 183 bytes of route-level JavaScript and halved its Firestore reads by rendering hand-built SVG charts as React Server Components, all covered by a 120-test Vitest suite in GitHub Actions CI

CipherWatch - Secret Exposure and Risk Detection | *Python, FastAPI, SQLAlchemy, WebSockets, APScheduler, React, TypeScript, pytest*

- Delivered an async FastAPI backend of 18 REST endpoints plus a WebSocket channel streaming 4 scan-lifecycle events, running scans as background tasks with APScheduler cron schedules and SMTP alerting, deployed at zero cost on Render and Vercel free tiers
- Bounded every input on an unauthenticated API (512 KB pastes, a GitHub URL allowlist, 1-1,000 commit caps, entropy bounds, capped page sizes) and closed a zip-slip path traversal in archive extraction plus two WebSocket registry leaks found in a codebase audit
- Cut SQL per dashboard-stats request from 9 to 6 by collapsing four per-severity counts into one GROUP BY and indexing the 5 hot filter and sort columns, and cut shipped JavaScript 56 percent (733 to 321 KB) by replacing charting and icon libraries with hand-built SVG

NeuroClass - Brain Tumor MRI Classifier (IEEE 2024) | *Python, ONNX Runtime, FastAPI, Docker, Gradio, Hugging Face, GitHub Actions*

- Served a calibrated MRI classifier at 33 ms CPU inference, 3.5x faster than Keras, by exporting to ONNX Runtime behind a FastAPI endpoint returning class, confidence, and Grad-CAM heatmaps, containerized with Docker and deployed on Hugging Face Spaces
- Rescued inference accuracy from 53 percent to 93.9 percent after a silent collapse toward one class, by isolating the fault to non-converged BatchNorm running statistics and re-estimating them with forward passes that leave the learned weights untouched
- Co-authored the underlying IEEE ICC-ROBINS 2024 publication, and kept the system reproducible and safe with a 10-stage one-command pipeline, 27 pytest tests gating CI, a versioned model card, and a 10 MB upload cap blocking memory-exhaustion attacks

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL

Backend and APIs: Node.js, Express.js, Fastify, FastAPI, REST APIs, WebSockets, Server-Sent Events, Zod, Pydantic

Frontend: React, Next.js, React Native, Tailwind CSS, React Query

Databases and Storage: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Firebase, pgvector, SQLAlchemy, Drizzle ORM

AI and ML: PyTorch, TensorFlow/Keras, Scikit-learn, Transformers, ONNX Runtime, RAG, LangGraph, QLoRA, Hugging Face

Cloud, DevOps and Testing: AWS (EC2, S3, Lambda, RDS), Docker, CI/CD (GitHub Actions), Git, Vercel, Vitest, pytest